

Seidel Wavefront Surrogate

A fast, differentiable model for optical system optimization

The Problem

Find surface shapes and arrangements that transform input wavefronts into desired output wavefronts.

- Point source $\rightarrow$ diverging sphere
- Perfect lens $\rightarrow$ converging sphere focused to a point
- Real lens $\rightarrow$ imperfect sphere
- **Wavefront error** = deviation from perfect sphere

What a Surface Does

Each refracting surface adds optical path length (OPL) that varies across the aperture:

$$\Delta W(x, y) = (n_2 - n_1) \cdot z(x, y)$$

- n_1, n_2 : refractive indices before/after
- $z(x, y)$: surface sag (shape)

For K surfaces, total wavefront error:

$$W_{\text{total}} = \sum_{i=1}^K (n'_i - n_i) \cdot z_i(y_i) - W_{\text{reference}}$$

The hard part: y_i at surface i depends on all previous surfaces.

Single Surface: Setup

Point source at distance s , spherical surface with curvature $c = 1/R$

Incoming wavefront (expand to 4th order):

$$W_{\text{in}}(y) = n_1 \left(\frac{y^2}{2s} + \frac{y^4}{8s^3} + \dots \right)$$

Surface sag (sphere Taylor expansion):

$$z(y) = \frac{cy^2}{2} + \frac{c^3y^4}{8} + \dots$$

Surface contribution:

$$\Delta W(y) = (n_2 - n_1) \left(\frac{cy^2}{2} + \frac{c^3y^4}{8} + \dots \right)$$

Single Surface: Two Orders

Setting $W_{\text{in}} + \Delta W = W_{\text{out}}$ order by order:

2nd order (y^2) — Paraxial imaging

$$\frac{n_1}{s} + (n_2 - n_1)c = \frac{n_2}{s'}$$

Determines image location. No aberration.

4th order (y^4) — Spherical aberration

$$W_{\text{error}} = \left(\frac{n_1}{8s^3} + \frac{(n_2 - n_1)c^3}{8} - \frac{n_2}{8s'^3} \right) y^4$$

Residual error. Edge rays suffer most. Cannot be zeroed independently for a single surface.

Multiple Surfaces: Degrees of Freedom

Surfaces	Free parameters	Can correct
1	0 (fixed by paraxial)	Nothing
2 (single lens)	1 (bending)	Minimize, not zero, S_I
5+	More than 5 Seidel types	Can zero all 3rd-order

The coupling problem: Changing surface 1 changes ray heights at surface 2, which changes surface 2's aberration contribution.

⇒ Nested nonlinear composition of coupled functions

The Seidel Simplification

Key move: Replace real ray height $y_i = h_i + \delta_i$ with paraxial ray height h_i

Drop the aberration-induced deviation δ_i .

Before (exact):

- Nonlinear nested composition
- Each surface depends on actual ray paths through all prior surfaces

After (Seidel):

- Linear chain (paraxial trace) — computed once
- Simple sum (aberration contributions from each surface)

This is what makes the model fast and differentiable.

The y - ν Trace: Variables

Track two quantities through the system:

$$y : \text{ray height} \quad \nu = n \cdot u : \text{reduced angle}$$

Using $\nu = n \cdot u$ instead of u makes Snell's law linear:

Refraction (at surface with power $\varphi_i = (n'_i - n_i) \cdot c_i$):

$$\nu' = \nu - y \cdot \varphi$$

Propagation (gap of thickness d in medium of index n):

$$y_{\text{next}} = y + \frac{d}{n} \cdot \nu$$

The y - ν Trace: The Chain

Alternating refract $\rightarrow$ propagate $\rightarrow$ refract $\rightarrow$ propagate:

$$\nu'_1 = \nu_1 - h_1 \cdot \varphi_1$$

$$h_2 = h_1 + \frac{d_1}{n'_1} \cdot \nu'_1$$

$$\nu'_2 = \nu'_1 - h_2 \cdot \varphi_2$$

$$h_3 = h_2 + \frac{d_2}{n'_2} \cdot \nu'_2$$

Changing $c_1 \rightarrow$ changes $\varphi_1 \rightarrow$ changes $\nu'_1 \rightarrow$ changes $h_2 \rightarrow$ ripples downstream.

Coupling is real but linear — much simpler than full raytracing.

Why Two Rays?

Aberrations depend on two independent dimensions:

Marginal Ray (aperture)

- Starts on-axis at object
- Goes to edge of aperture stop
- h_i = beam footprint at surface i

Chief Ray (field)

- Starts at edge of field of view
- Passes through center of stop
- $\bar{h}_i$ = off-axis beam offset at surface i

Both traced with the same y - ν recurrence.

At each surface we get: h_i, ν_i (marginal) and $\bar{h}_i, \bar{\nu}_i$ (chief)

Seidel Coefficients: Building Blocks

At each surface i , compute:

Refraction invariants:

$$A_i = \nu_i + n_i \cdot h_i \cdot c_i$$

$$\bar{A}_i = \bar{\nu}_i + n_i \cdot \bar{h}_i \cdot c_i$$

Refraction change:

$$\Delta(u/n)_i = \frac{u'_i}{n'_i} - \frac{u_i}{n_i}$$

Lagrange invariant (constant through system):

$$H = n_i \cdot u_i \cdot \bar{h}_i - n_i \cdot \bar{u}_i \cdot h_i$$

The Five Seidel Aberrations

$$S_{I,i} = -A_i^2 \cdot h_i \cdot \Delta(u/n)_i$$

$$S_{II,i} = -A_i \cdot \bar{A}_i \cdot h_i \cdot \Delta(u/n)_i$$

$$S_{III,i} = -\bar{A}_i^2 \cdot h_i \cdot \Delta(u/n)_i$$

$$S_{IV,i} = -H^2 \cdot \frac{\varphi_i}{n_i \cdot n'_i}$$

$$S_{V,i} = (S_{III,i} + S_{IV,i}) \cdot \frac{\bar{A}_i}{A_i}$$

Aperture-Field Structure

Each aberration has a distinct dependence on aperture (h) and field ($\bar{h}$):

Aberration	Dependence	Character
Spherical	h^4	Pure aperture
Coma	$h^3\bar{h}$	Mostly aperture
Astigmatism	$h^2\bar{h}^2$	Equal mix
Field curvature	$h^2\bar{h}^2$	Equal mix (different symmetry)
Distortion	$h\bar{h}^3$	Mostly field

This is why we need **both** rays.

Summing Over Surfaces

Total system aberrations — simple sums:

$$\begin{aligned} S_I &= \sum_i S_{I,i} & S_{II} &= \sum_i S_{II,i} & S_{III} &= \sum_i S_{III,i} \\ S_{IV} &= \sum_i S_{IV,i} & S_V &= \sum_i S_{V,i} \end{aligned}$$

Each term shares the y - ν chain $\Rightarrow$ not independent w.r.t. design variables.

But the sum itself is trivial once the trace is done.

Aspheric Surfaces

Sag with conic k and polynomial A_4 :

$$z(y) = \frac{cy^2/2}{1 + \sqrt{1 - (1 + k)c^2y^2}} + A_4y^4 + A_6y^6 + \dots$$

Extra contributions to spherical aberration:

$$S_{I,\text{conic}} = -(n'_i - n_i) \cdot k_i \cdot c_i^3 \cdot h_i^4$$

$$S_{I,A_4} = -8(n'_i - n_i) \cdot A_{4,i} \cdot h_i^4$$

Aspherics don't affect the paraxial trace — they only add to aberration sums.

Aspherics: Stop Position Matters

Away from the stop ($\bar{h} \neq 0$), aspherics also correct off-axis aberrations:

$$S_{II,\text{asph}} = S_{I,\text{asph}} \cdot \frac{\bar{h}_i}{h_i}$$

$$S_{III,\text{asph}} = S_{I,\text{asph}} \cdot \left(\frac{\bar{h}_i}{h_i} \right)^2$$

Aspheric location	Corrects
At the stop ($\bar{h} = 0$)	Spherical only
Away from stop ($\bar{h} \neq 0$)	Spherical + Coma + Astigmatism

Key design lever for multi-aberration correction.

Wavefront Error Expansion

Total wavefront as function of pupil ρ and field η :

$$W(\rho, \eta, \theta) = S_I \rho^4 + S_{II} \eta \rho^3 \cos \theta + S_{III} \eta^2 \rho^2 \cos^2 \theta \\ + S_{IV} \eta^2 \rho^2 + S_V \eta^3 \rho \cos \theta$$

Maps to Zernike polynomials:

Zernike	Aberration	From
a_{11}	Spherical	S_I
a_7, a_8	Coma	S_{II}
a_5, a_6	Astigmatism	S_{III}
a_4	Field curvature	$S_{III} + S_{IV}$

Merit Function

Weighted sum of squared aberrations over field points:

$$M = \sum_{\text{field points}} \sum_j w_j \cdot S_j^2$$

Weights encode design priorities:

- On-axis system $\rightarrow$ weight S_I heavily
- Wide-field system $\rightarrow$ weight S_{III}, S_{IV} heavily
- Imaging system $\rightarrow$ balance all five

Gradients: Why This Model is Fast

Every S_j is an explicit polynomial function of design variables $(c_i, d_i, k_i, A_4, n_i)$.

Dependency chain:

$$c_i, d_i \xrightarrow{\text{y-}\nu \text{ trace}} h_i, \nu_i \xrightarrow{\text{invariants}} A_i, \bar{A}_i \xrightarrow{\text{Seidel}} S_j \xrightarrow{\text{square}} M$$

Chain rule gives exact gradients:

$$\frac{\partial M}{\partial c_1} = 2 \sum_j w_j \cdot S_j \cdot \frac{\partial S_j}{\partial c_1}$$

Cost: $O(K \times V)$ — same order as one forward evaluation.

No finite differences. No raytracing. Pure arithmetic.

Validity

Captures

- All five 3rd-order monochromatic aberrations
- Correct surface-by-surface contributions and field dependence
- Aspheric effects (conic + polynomial)
- Linear coupling through y - ν chain

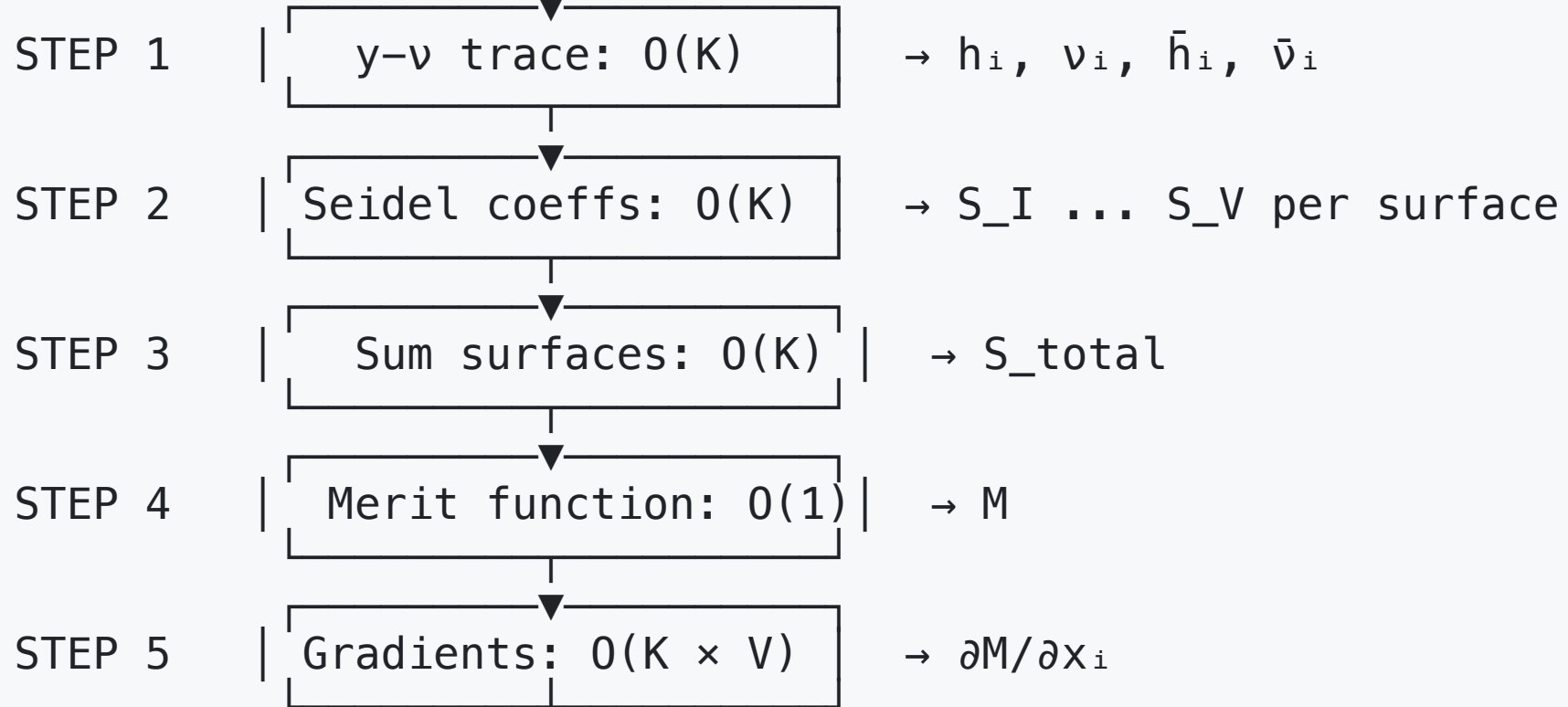
Misses

- Higher-order aberrations (5th, 7th) — matters below $\sim f/2$
- Chromatic aberrations (unless multi-wavelength trace)
- Nonlinear ray height coupling (very steep rays)
- Vignetting

Reliable for $f/2$ and slower. Finds correct optimization basins even when absolute values

Computational Pipeline

INPUT: Prescription (c, d, n, k, A_4) + Specs (object dist, field, aperture)



TOTAL: $O(K \times V)$ per evaluation – pure arithmetic

Design Variables & Constraints

Variables

- c_i : curvatures
- d_i : spacings / thicknesses
- k_i : conic constants
- A_4, A_6 : aspheric coefficients
- n_i : refractive indices

Constraints

- Focal length (paraxial power)
- $d_i > 0$ (physical thickness)
- Surfaces must not intersect
- Glass must exist in catalog

All continuous except glass selection (discrete $\rightarrow$ continuous relaxation, match later).

Summary

1. **Physical model:** wavefront error = sum of OPL contributions from each surface
2. **Seidel approximation:** use paraxial ray heights, decouple the nonlinear nesting
3. **y- ν trace:** linear chain giving ray data at each surface
4. **Five aberrations:** computed from trace data via explicit formulas
5. **Merit function:** weighted sum of squares — smooth, differentiable
6. **Gradients:** exact, cheap, no raytracing needed

Full optimization loop: $O(K \times V)$ arithmetic operations